MIL-F-36933A
30 August 1977
SUPERSEDING
MIL-F-36933
16 January 1973

MILITARY SPECIFICATION

FORCEPS, HEMOSTATIC, HALSTED, 5 INCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers curved and straight hemostatic forceps of the Halsted design.

1.2 Classification. Forceps shall be of the following types as indicated below (see 6.1).

Type

1 - Curved.
2 - Straight.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

GG-I-526 - Instruments, Dental and Surgical; General Specifications.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Set-up.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:36Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-L-10547 - Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

20251 - FORCEPS, HEMOSTATIC, HALSTED.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests for.

E 92 - Vickers Hardness of Metallic Materials, Tests for.

E 140 - Hardness Conversion Tables for Metals (Relationship Between Brinell Hardness, Vickers Hardness, Rockwell Hardness, Rockwell Superficial Hardness and Knoop Hardness).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:36Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General requirements. The hemostatic forceps shall be in accordance with the applicable requirements of GG-I-526, and as specified herein.

3.2 Material. The forceps shall be fabricated from Class 4, type 410 or 420 material.

3.3 Hardness. After proper stress relieving, Rockwell hardness shall be C40 to C45 when tested as specified in 4.4.1. The requirements of GG-I-526 shall apply.

3.4 Style, design and dimensions. The forceps shall be mosquito type of the Halsted design. Both forceps, straight and curved types, specified herein, shall be suitable for accomplishing hemostasis in delicate tissue. Both jaws of the forceps shall be serrated throughout their entire length. Serrations shall be consistent in depth and in accordance with GG-I-526. The forceps shall mesh completely and properly during closure. The action of the forceps shall be easy, without drag or binding action. The forceps shall be in strict accordance with Defense Personnel Support Center (DPSC) Drawing 20251 for the appropriate type and National Stock Number (NSN).

3.5 Locks. Shall be of box lock type. The box lock construction shall be accurately fitted, and shall be without crevices, burrs, or sharp edges at any place in the lock construction. Pin or rivet shall be permanently secured. The joint performance shall be smooth, of equal resistance, and non-binding in all positions when opening or closing the forceps (the lock may be lubricated). Further, under no circumstances shall looseness or free play of either shank permit any misalignment of the jaws in closing. NOTE: This paragraph shall replace the requirements specified in GG-I-526 for Locks.

3.6 Finish. Finish shall be entirely mirror, satin or matte. Should the instrument be of mirror finish, the locks and ratchets shall be polished smooth, but need not possess a mirror finish. The matte finish shall exhibit a uniform nondirectional surface texture equivalent to that produced by lightly blasting the prepared surface with fine abrasives under pressure.

3.6.1 Passivation. The forceps shall be properly passivated.

3.7 Jaw and ratchet operation. When tested as specified in 4.4.2, the jaw tips shall close first and progressively tighten with each ratchet position. Each ratchet position shall hold firmly and release instantly.

3.8 Identification marking. In addition to the requirements specified in GG-I-526, forceps of foreign manufacture shall be marked "stainless".

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:36Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection shall be in accordance with GG-I-526.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the corrosion-resistant steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table I. The unit of product for examination purposes shall be one forceps.

TABLE I. Sampling criteria for inspection.

Type defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in table II. The unit of product for test purposes shall be one forceps. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:36Z
Check the source to verify that this is the current version before use.

TABLE II. Sampling criteria for tests.

Component	Characteristic	Require-ment	Test procedure	Inspection level	AQL (percent defective)
Forceps	Hardness test	3.3 & GG-I-526	4.4.1	S-2	1.5
	Dimensions	3.4	----	S-2	2.5
	Distortion test	GG-I-526	GG-I-526	S-2	2.5
	Copper sulfate test	GG-I-526	GG-I-526	S-2	1.0
	Boil test	GG-I-526	GG-I-526	S-2	1.0
	Jaw and ratchet operation test	3.7	4.4.2	S-2	1.0

4.2.3 Raw materials. A sufficient quantity of each lot of metal used in the manufacture of the forceps shall be selected for chemical analysis.

4.3 Examination.

4.3.1 Examination of forceps. The forceps shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any forceps in the sample containing one or more defects shall be rejected and if the number of defective forceps in any sample exceeds the acceptance number for the sample, the lot represented by that sample shall be rejected.

TABLE III. Classification of defects.

Categories	Defects
Critical	None defined.
Major	
101	Forceps not of the specified design for appropriate type, stock number and item identification.
102	Both jaws not serrated throughout full length.
103	Jaw serrations not of uniform depth and not well defined.
104	Jaw serrations not chamfered throughout full length.
105	Ratchet teeth not well defined.
106	Forceps not free from cracks, pits, burrs, sharp edges, scale, rough areas or corrosion.
107	Finish of forceps not uniform and not as specified.
108	Serrations of both jaws do not completely mesh in fully closed position.
109	Box lock performance not as specified in 3.5.
110	Jaws possess sharp tips.
111	Ratchet teeth do not completely engage in all positions.
112	Pin for box lock not flush with surface and not free from crevices.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:36Z
Check the source to verify that this is the current version before use.

TABLE III. Classification of defects.- continued

Categories	Defects
Minor	
201	Forceps not free from dirt, polishing material or foreign matter.
202	Identification markings not present, not correct, not complete, or not legible.

4.3.2 Packaging inspection. The inspection of the preservation-packaging shall be in accordance with group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Hardness test. Hardness shall be determined in accordance with ASTM Standards E 18 or E 92. Conversion of Vickers hardness values to Rockwell values shall be in accordance with ASTM Standard E 140.

4.4.2 Jaw and operation test. The forceps shall be positioned to the first ratchet position, but not engaged. Forceps with jaw tips not touching shall be rejected; further, should all the jaw serrations touch, the forceps shall be rejected. The forceps shall then be engaged to each of the ratchet positions. Forceps that do not progressively tighten with each ratchet advance shall be rejected.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Each forceps shall be cleaned and dried with the applicable method as outlined in MIL-P-116.

5.1.1.2 Preservative. Preservative shall not be applied.

5.1.1.3 Unit packaging. Each forceps shall be wrapped in neutral tissue paper or other suitable anticorrosive material. Each forceps shall be further packaged in accordance with MIL-P-116, method 1C-1.

Source: https://assist.dla.mil -- Downloaded: 202610-21T09:36Z
Check the source to verify that this is the current version before use.

5.1.1.4 Intermediate package. Twelve (12) forceps shall be packaged in a box of appropriate size constructed in accordance with PPP-B-566, PPP-B-676 or PPP-B-636, type CF, class domestic. Closure of box shall be as specified in the appendix of the applicable box specification.

5.1.2 Level C. Forceps shall be clean and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Three hundred and sixty (360) forceps shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2 or PPP-B-636, class weather resistant, grade V3c. Closure and shipping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof as specified in the appendix of PPP-B-636.

5.2.2 Level B. Three hundred and sixty (360) forceps shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure of box shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The packaged forceps shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carriers rules and regulations.

5.3 Marking. Each unit, intermediate package, and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Applicable NSN.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:36Z
Check the source to verify that this is the current version before use.

6.2 National Stock Number coverage. This specification covers the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identification	Specification Designation
6515-00-334-4900	FORCEPS, HEMOSTATIC, Curved, Halsted, 5 inch.	Type 1
6515-00-334-5600	FORCEPS, HEMOSTATIC, Straight, Halsted, 5 inch.	Type 2

6.3 Matte and satin finish. It should be noted that matte finish specified herein can be distinguishable from mirror and satin finishes under high-power illuminated magnification. Forceps of matte finish shall have a reduced reflective surface that is more glare resistance than satin finish.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Reviewing activity:
DOD - MB

Preparing activity:
Defense Logistics Agency - DM

Project No. 6515-2067

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:36Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE PERSONNEL SUPPORT CENTER
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

POSTAGE AND FEES PAID

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 202640-21T09:36Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS. This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER (Number) AND TITLE

MIL-F-36933A FORCEPS, HEMOSTATIC, HALSTED, 5 INCH

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY (Printed or typed name and address — Optional)

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:36Z
Check the source to verify that this is the current version before use.